



## Project 0: Virtual Machine



### Learning Objectives

#### Phase 1: Cloud Foundation

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##### Task 2: Single Instance Deployment

#### Phase 2: High Availability

##### Task 3: Custom Machine Image

##### Task 4: Load Balancing & Redundancy

##### Task 5: Testing Resilience

##### Task 6: Cost Analysis & Cleanup

#### Phase 3: Automated Elasticity (Graduate Students Only)

##### Task 7: Auto Scaling & Self-Healing

### Submission Guidelines

### Grading Scheme

# Project 0: Virtual Machine

**Release Date:** Tuesday, February 3, 2026

**Due Date:** Tuesday, February 17, 2026, 11:59 PM EST

In this project, you will gain hands-on experience with Infrastructure-as-a-Service (IaaS). Rather than following a rigid set of instructions, you are tasked with exploring the cloud platform of your choice to architect a resilient, highly available web service.

You may utilize **Amazon Web Services (AWS)**, **Microsoft Azure**, or **Google Cloud Platform (GCP)**.

The core concepts—Virtual Machines, Networking, Load Balancing, and High Availability—are universal.

**Duration:** 2 weeks

### Learning Objectives

1. **Cloud Foundation:** establishing secure access and m
2. **High Availability:** implementing load balancing and n

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3. **Observability:** monitoring system health and verifying failover capabilities.
4. **Elasticity (Graduate Only):** implementing automated scaling and self-healing infrastructure.

## Phase 1: Cloud Foundation

Your first objective is to establish a working environment and launch a single server.

### Task 1: Environment Setup

- **Account Creation:** Sign up for your chosen cloud provider (ensure you use a free tier if available).
- **Security:** Enable Multi-Factor Authentication (MFA) for your root account. Create a restricted user (IAM User/Service Principal) for daily tasks; do not use the root account for resource creation.
- **CLI Configuration:** Install and configure the Command Line Interface (CLI) tool for your platform (e.g., `aws`, `az`, or `gcloud`). Verify you can query the cloud provider from your local terminal.

### Task 2: Single Instance Deployment

- **Network Security:** configure a firewall (Security Group) that allows:
  - **SSH (Port 22):** Only from your specific IP address.
  - **HTTP (Port 80):** From anywhere (0.0.0.0/0).
- **Compute:** Launch a Virtual Machine (e.g., EC2, Azure VM, Compute Engine) using a standard Linux distribution (Amazon Linux, Ubuntu, etc.).
- **Web Server:** Connect to your instance via SSH and install a web server (Apache or Nginx).
- **Content:** Configure the index page to display a custom “Hello World” message that includes the **hostname** or **internal IP** of the server.

**Verification:** You should be able to load the webpage using the VM’s public IP/DNS.

## Phase 2: High Availability

Now that you have a single server, your goal is to make the architecture resilient to failure.

### Task 3: Custom Machine Image

- Instead of manually installing software on every server, create a **Machine Image** (AMI/Image) from your running instance in Task 2.
- This image will serve as the template for your scalable fleet.

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## Task 4: Load Balancing & Redundancy

- **Infrastructure:** Provision an **Application Load Balancer** (L7).
- **Target Group:** Create a target group that performs health checks on your web server path.
- **Scaling:** Launch at least **two (2)** new instances using your Custom Image from Task 3.
  - **Crucial:** These instances must be placed in **different Availability Zones** (e.g., `us-east-1a` and `us-east-1b`) to ensure physical redundancy.
- **Routing:** Register these instances with your Load Balancer.

## Task 5: Testing Resilience

- **Traffic Distribution:** Continually refresh your Load Balancer's DNS URL. You should see the content change (switching between hostnames/IPs) as the traffic is distributed.
- **Simulated Failure:** Manually terminate (delete) one of the instances.
- **Verification:** Continue refreshing the Load Balancer URL. The service should remain online, serving traffic only from the remaining healthy instance.

## Task 6: Cost Analysis & Cleanup

- **Analysis:** Use your cloud provider's cost exploration tools or pricing calculator to estimate the monthly cost of this architecture if it were left running 24/7.
- **Cleanup:** Delete all resources (Load Balancers, Instances, Security Groups, Images, Snapshots) to avoid unexpected charges.

## Phase 3: Automated Elasticity (Graduate Students Only)

**Undergraduate students may attempt this for extra credit (up to +10 points). Graduate students are REQUIRED to complete this section.**

## Task 7: Auto Scaling & Self-Healing

Instead of manually launching instances (as done in Task 4), you must implement an **Auto Scaling Group** (AWS), **Virtual Machine Scale Set** (Azure), or **Managed Instance Group** (GCP).

- **Launch Template:** Create a configuration that utilizes the Custom Image created in Task 3.
- **Group Configuration:** Configure the group to maintain a **Desired Capacity of 2**.
- **Integration:** Attach this group to your Load Balancer Target Group.
- **Self-Healing Test:** Terminate an instance manually. Prove that the Auto Scaling Group detects the unhealthy/missing instance and **automatically launches a replacement without human intervention**.

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# Submission Guidelines

**Due Date:** Tuesday, February 17, 11:59 PM.

**Format:** Submit a **SINGLE PDF** file named `Project0_YourName.pdf` to Brightspace. Include your name and if you are a graduate student on the first page.

**Report Requirements:** Your report is the primary evidence of your work. For **EACH** task listed above, you must include:

1. **Commands Run:** The exact CLI commands you executed.
2. **Output:** The text output from those commands (e.g., JSON response showing instance ID, Running state) or screenshots where applicable.
3. **Configuration/Scripts:** Copy and paste the content of any bootstrap scripts (User Data) or policy files you used directly into the PDF.
4. **Explanation:** A brief paragraph explaining *why* you ran those commands or configured the architecture that way.
5. **Screenshots:** Visual proof is required for:

◦ The web page loading via the Load Balancer.

◦ The console showing instances in multiple Availability Zones.

◦ The “Simulated Failure” test.

◦ **(Graduate Only):** Screenshot of the Auto Scaling Group history/activity log showing the automatic replacement of a terminated instance.

# Grading Scheme

The project is graded out of **100 points**.

| Category                 | Points | Criteria                                                                                                                                                                                                                              |
|--------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documentation & Evidence | 25     | <b>Did you show your work?</b><br>Full points if the report includes the CLI commands run and their output/screenshots for every task. Deductions for missing logs, vague descriptions like “I clicked the button,” or missing steps. |
| Technical Correctness    | 25     | <b>Is the architecture valid?</b><br>Full points for correctly configured Security Groups, Custom Images, and Load Balancers. Deductions for incorrect IP access (0.0.0.0/0) or missing rules. <div>Chat Assistant</div>              |

| Category             | Points | Criteria                                                                                                                                                                                                                                                                                                                     |
|----------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concept Mastery      | 20     | <b>Do you understand what you did?</b><br>Points awarded for the <b>Explanations</b> in your report. <ul style="list-style-type: none"><li>• <i>Excellent</i>: Clear reasoning for Multi-AZ placement, health checks, and image creation.</li><li>• <i>Poor</i>: No explanation, or fundamental misunderstandings.</li></ul> |
| Execution            | 15     | <b>Did it actually work?</b><br>Evidence (screenshots) showing the app running via the Load Balancer, demonstrating traffic distribution (different Hostnames), and surviving instance termination.                                                                                                                          |
| Graduate Requirement | 15     | <b>(Required for Grads / Bonus for Undergrads)</b><br>Successful implementation of <b>Auto Scaling</b> . Points awarded for evidence of the system automatically replacing a terminated instance (Self-Healing).<br><i>Note: Undergrads not attempting this are graded out of 85 and normalized to 100.</i>                  |

**Common Deductions:**

- **-10 points**: Submitting separate files/images instead of a single PDF.
- **-10 points**: Missing the "Hostname/IP" on the webpage (which proves load balancing is actually distributing traffic).
- **-15 points**: Instances launched in the **same** Availability Zone (fails the High Availability requirement).
- **-50 points**: Leaking AWS/Azure Access Keys or Secrets in your PDF report (Security Risk).

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